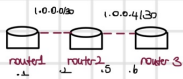


For routers:

192.168.10.0
255.255.255.0



network: 1.0.0.0/30
broadcast: 1.0.0.3

network: 1.0.0.4/30
broadcast: 1.0.0.7/30

A) 31 hosts:

192.168.10.0

$2^6 - 2 = 62$ hosts

255.255.255.11000000
.192 or /26

netmask $\rightarrow$ /26

network address 192.168.10.0/26

first host address 192.168.10.1

broadcast address 192.168.10.63

00:00 00 00
11 11 11
-63 $\rightarrow$ broadcast

B) 10 hosts

192.168.10.64

$2^4 - 2 = 14$ hosts

255.255.255.11110000
.240 or /28

netmask: /28

network address: 192.168.10.64/28

first host address: 192.168.10.65/28

broadcast address: 192.168.10.79

0100:0000
1111
-79 $\rightarrow$ broadcast

C) 10 hosts:

192.168.10.80

$2^4 - 2 = 14$ hosts

255.255.255.11110000

.240 or /28

netmask: /28

network address: 192.168.10.80/28

first host address 192.168.10.81/28

broadcast address 192.168.10.95/28

0101:0000
1111
-95 $\rightarrow$ broadcast

D) 2 hosts

192.168.10.96

$2^2 - 2 = 2$ hosts

255.255.255.11111100

.252 or /30

netmask: /30

network address 192.168.10.96/30

first host address 192.168.10.97/30

broadcast address 192.168.10.99/30

011000:00
11
-99 $\rightarrow$ broadcast